



OCT 2024



Department of Chemical Engineering

Vikas Berry

Department Head and Satish C. & Asha Saxena Professor





DEPARTMENT UPDATES



43rd Academic Year as Chemical Engineering at UIC



STUDENTS AND FACULTY

Faculty Members : 16 (13.25 FTE)
National Award Winners // CEOs // Industry Experts

Tenure-Track : 11
Full : 4
Associate : 5
Assistant : 2
% Female : 18.18%
Bridge-2-Faculty: 1

Clinical-Track
Assistant : 1
Associate : 3
% Female : 25%

Research-Track : 1

Affiliated Faculty : 18

BS Degree Program : 165 students
Graduate Program : 55 students
MS Degree Program : 13 students
PhD Degree Program : 42 students
BS-MS Joint Program : 10 students

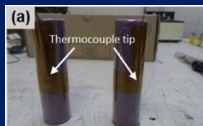
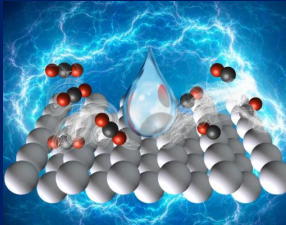
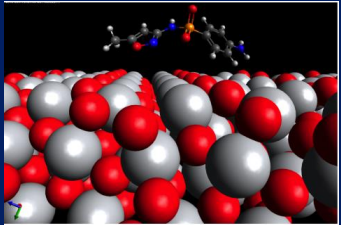
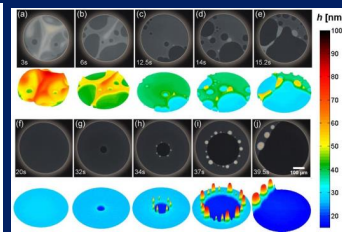
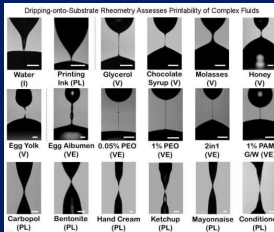
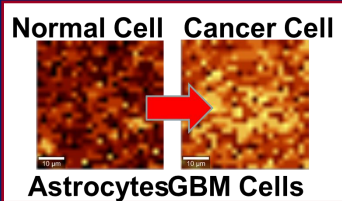
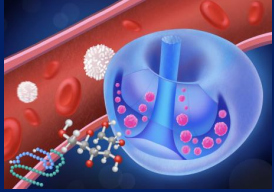


Diversity Statement

UIC chemical engineering fosters an inclusive environment that encourages, supports, and empowers diverse voices, ideas, and opinions. We value equity, inclusion, and all forms of diversity within our faculty, staff, and student population. Our multicultural, pluralistic community brings richness in experience, backgrounds, perspectives, and skills to all activities in the department.



STAFF



Sarai Chavez

– Office Support Assistant

Meghan King

– Assistant Director of Academic Communications

Eduardo Martinez

– Grad./Undergraduate Program Coordinator

Jan Raphael Sagun

– Laboratory and Computers Manager

Cassandra Wilson

– Business Manager

COE (Monica Tith)

– HR



DEPARTMENT UPDATES



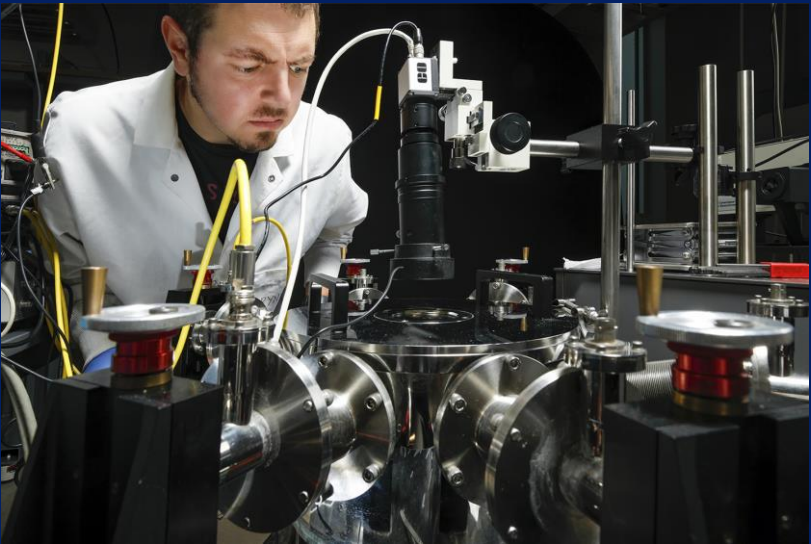
Rankings

- **US-NEWS**
 - Chemical Engineering (overall) = 62
 - Chemical Engineering (public) = 38
- **QS Ranking**
 - Chemical Engineering (overall) = 35
 - Chemical Engineering (public) = 22

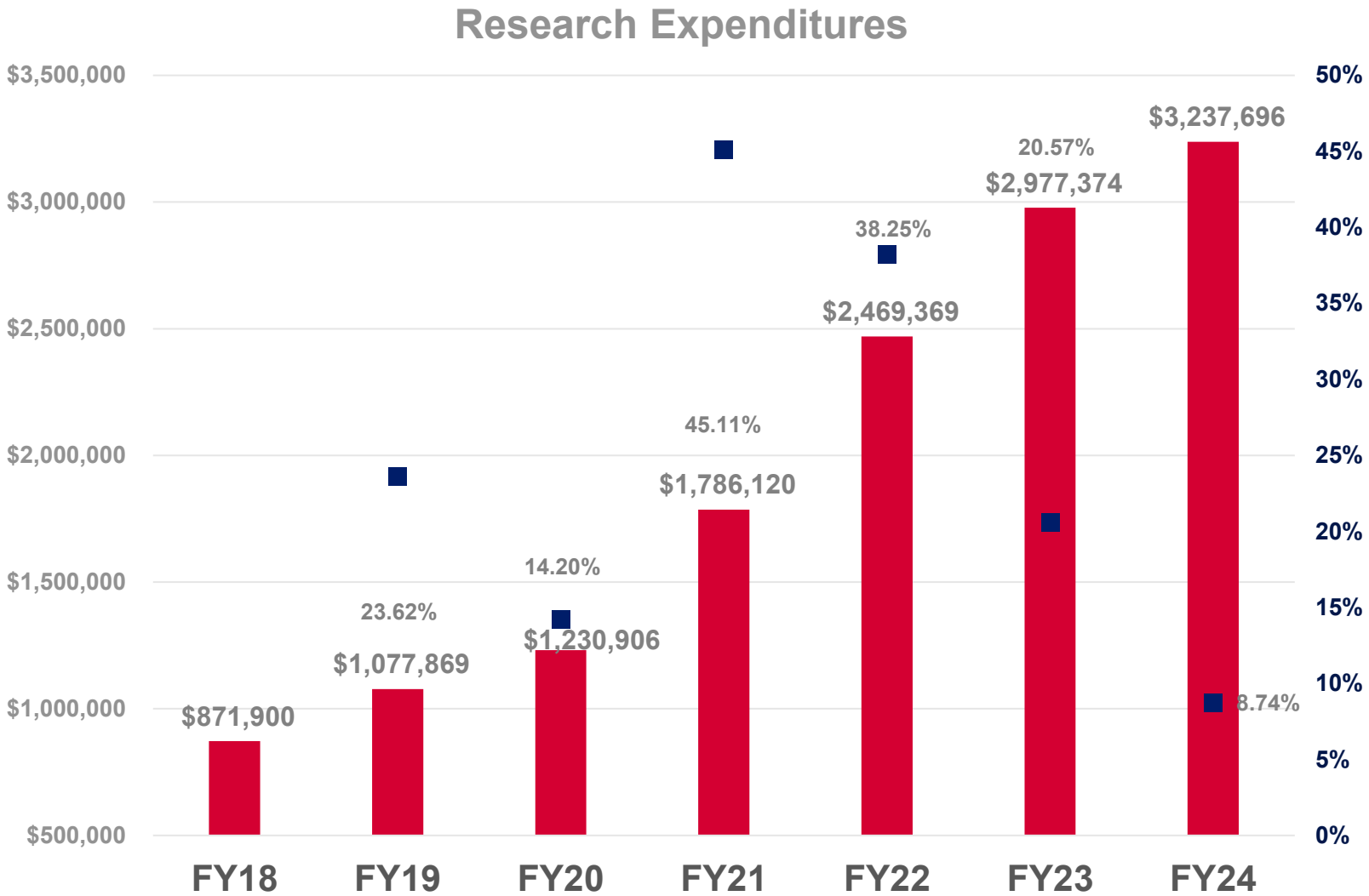


DEPARTMENT UPDATES

Expenditures



Research Expenditures:



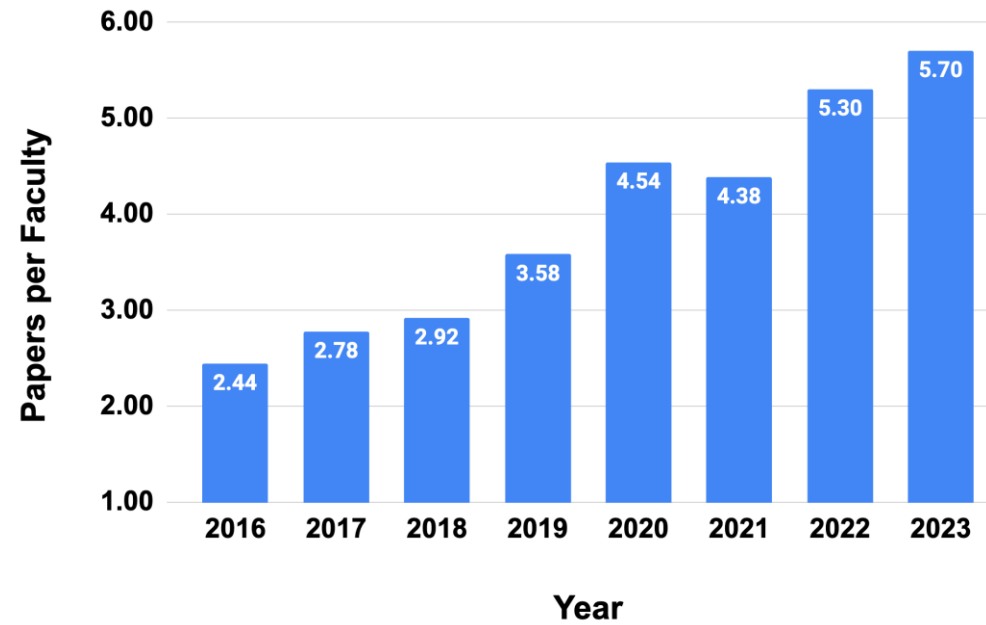


RESEARCH PRODUCTIVITY

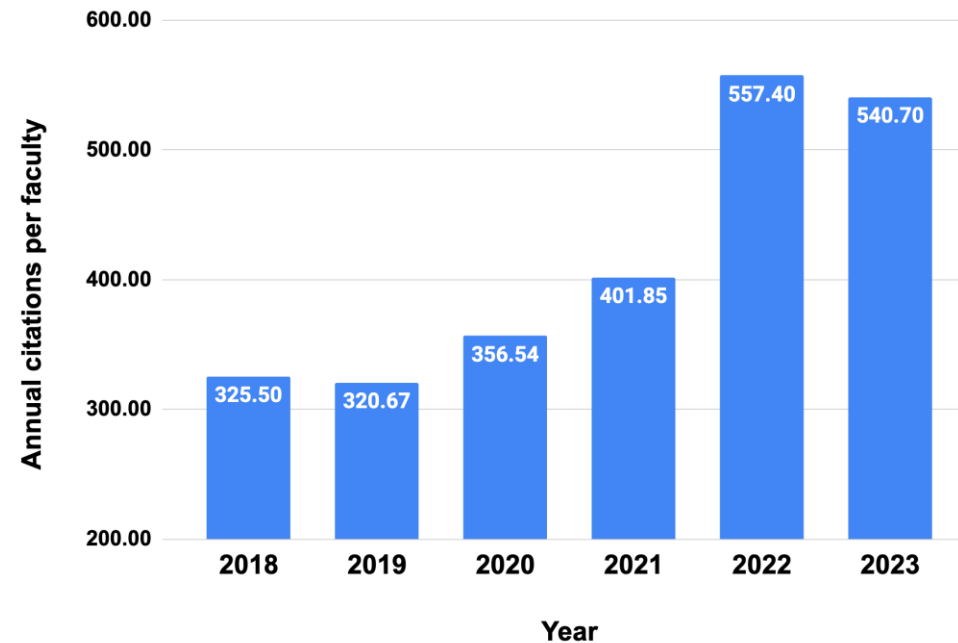
Publications

Tot Citations/Fac = 3971

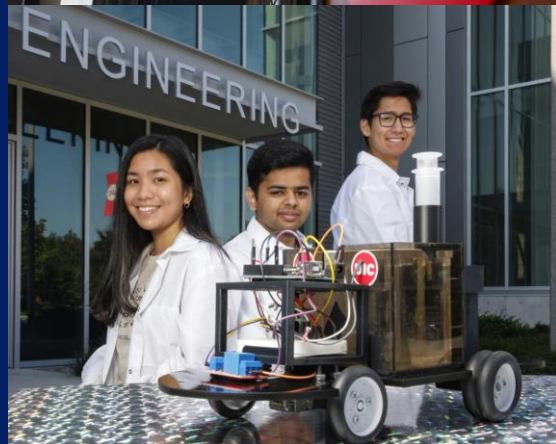
Avg. h-Index = 24.31



Average Citations



KEY ACCOMPLISHMENTS



STUDENT PLACEMENTS



L'ORÉAL



Exxon

TESLA



JACOBS



CABOT

GELITA
Improving Quality of Life

LA-CO



Ambitech

GATX



NAVISTAR



ENEngineering

EXIDE
TECHNOLOGIES



South
Chicago
Packing



Air Liquide

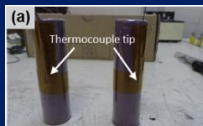
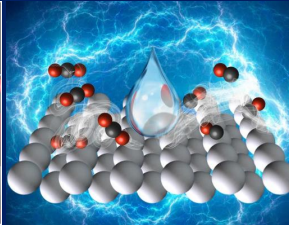
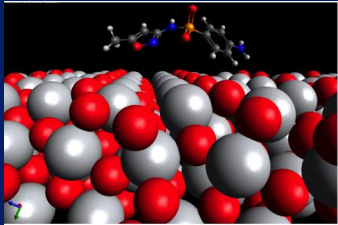
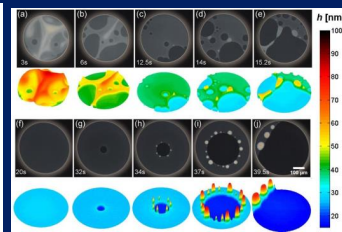
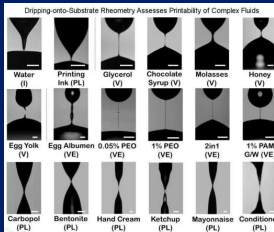
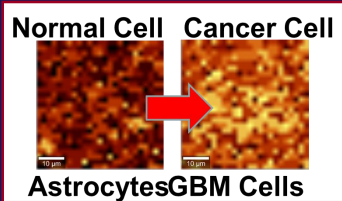
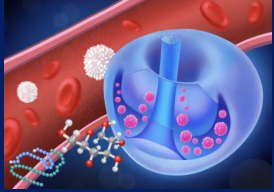


middough





KEY ACCOMPLISHMENTS



Chemical Engineering is Sustainability

- Energy
 - Battery Research
 - Sustainable Energy Technologies
- Water
 - Water Resource Sustainability
- Human Health
 - Therapeutics (Drug Delivery)
 - Diagnostics
- Food
- Materials
- Funding in force: \$15M

The Stanford Top 2% researchers rankings report

WORLD-CLASS
FACULTY MAKES A
WORLD-CLASS
DEPARTMENT



Career rankings from 1960:

- Professor Vikas Berry
- Professor Brian Chaplin
- Research Professor Said Al-Hallaj

Citations and publications for 2022:

- Professor Brian Chaplin
- Professor Vikas Berry
- Research Professor Said Al-Hallaj
- Associate Professor Meenesh Singh
- Associate Professor Vivek Sharma

**WORLD-CLASS
FACULTY MAKES A
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Entrepreneurship

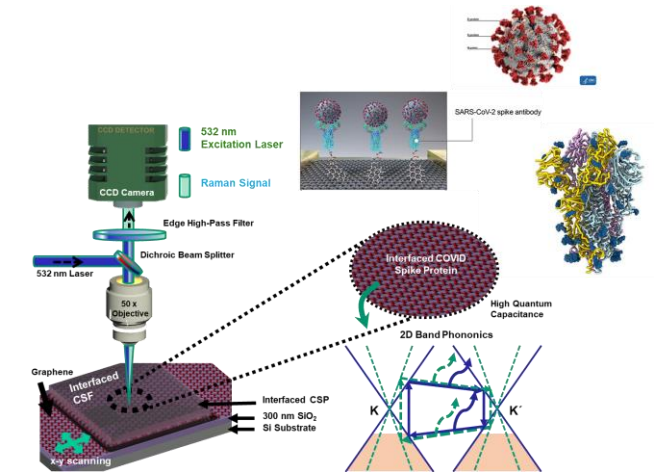
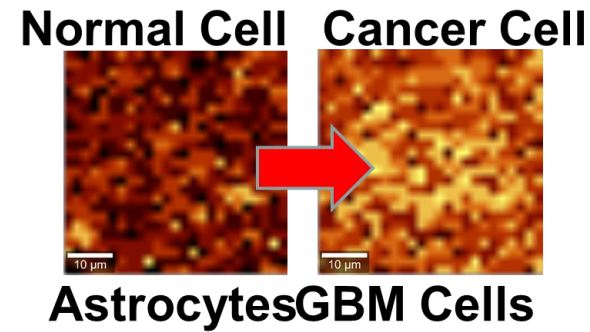
Chemical Engineering Faculty and Students are collaborating with the following resources to translate ideas into products



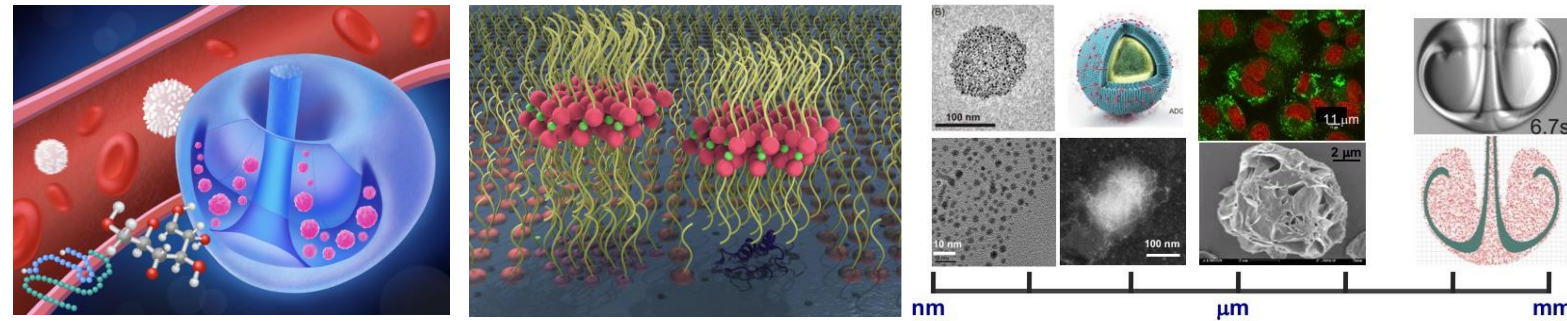
13 faculty/student startups
3 CTRI grants received
2 POC grants received
1 DPI grant received
3 NSF-ICORPs
2 operations in mHUB

SOME RESEARCH AREAS

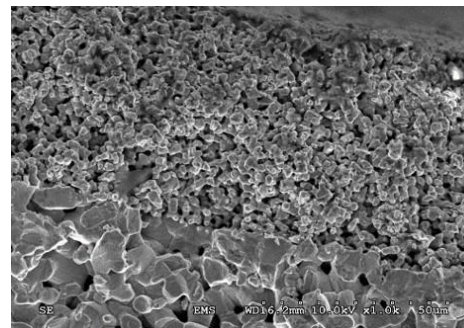
- COVID and Cancer Detection



- Drug Delivery for Vaccine Development



- Water



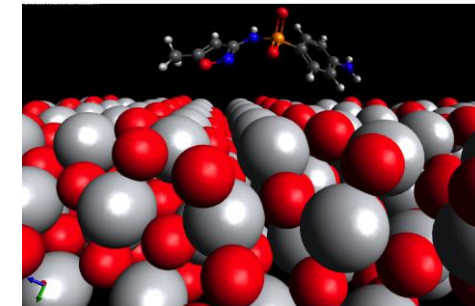
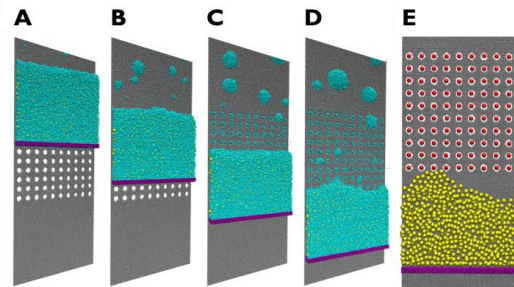
Engineering/Chemical Engineering/

SOME RESEARCH AREAS

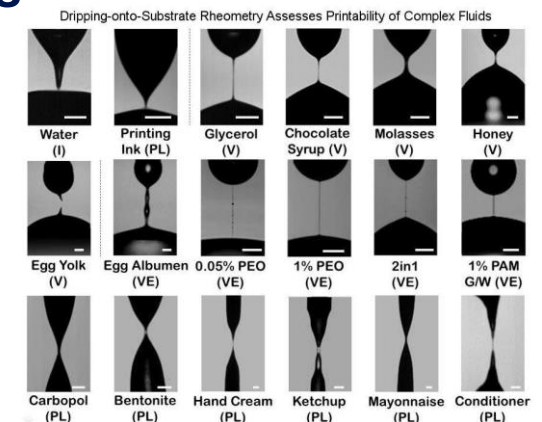
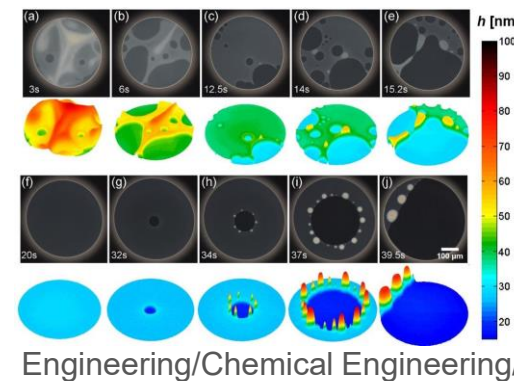
- Bioenergy and Sustainability



- Simulation of Self-Assembly in Nanotechnology

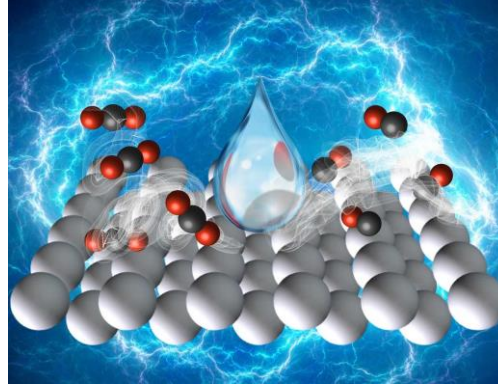


- Printability of Complex Fluids

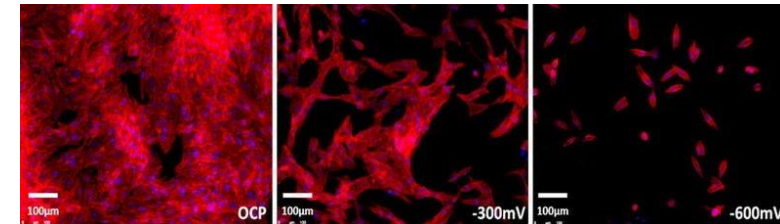


SOME RESEARCH AREAS

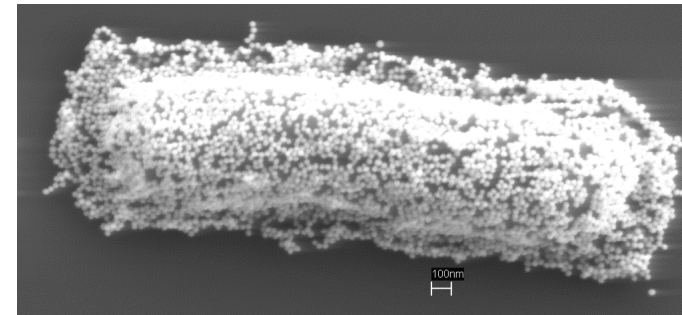
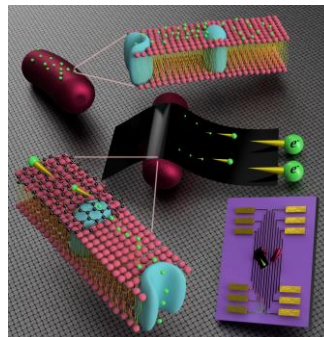
- **Artificial Photosynthesis, Crystallization**



- **Microelectronics Processing**



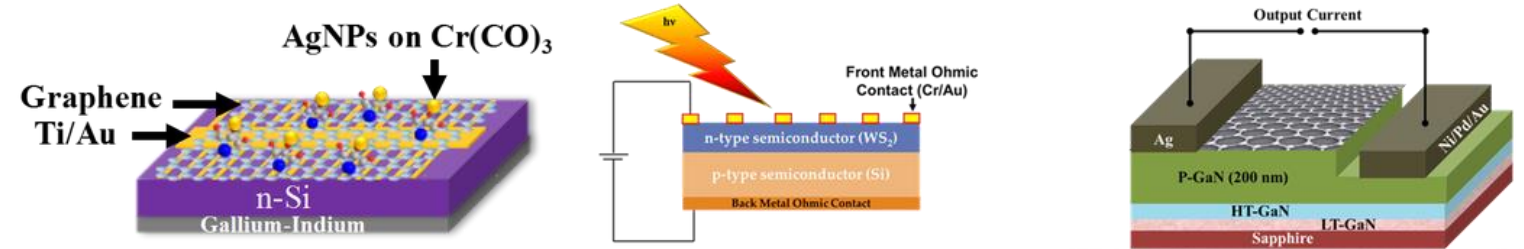
- **Microbial Fuel Cells**



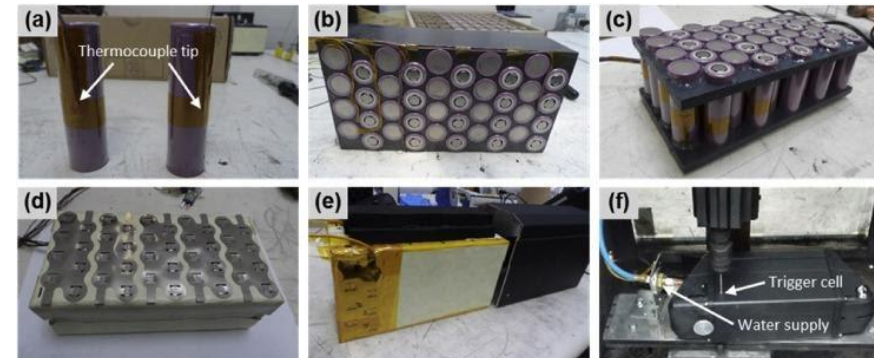
Engineering/Chemical Engineering/

SOME RESEARCH AREAS

- **Few-Atoms-Thick Solar Cells**



- **Thermal Batteries**





OCTOBER 1, 2024

Town Hall Meeting: ChE Undergraduate Program

Alan Zdunek
Clinical Associate Professor
Director of Undergraduate Studies



UNIVERSITY OF
ILLINOIS CHICAGO

ABET Program Educational Objectives

PEO	Category	Description
1	Depth	In engineering practice, advanced study or original research, UIC Chemical Engineering bachelor's degree graduates will be effective in applying fundamental principles, scientific knowledge, rigorous analysis, creative design, and conceptual innovation.
2	Breadth	Based upon mastery of engineering in a broad, societal context, UIC Chemical Engineering graduates will have successful careers in the public or private sectors, or in the pursuit of graduate education.
3	Professionalism	In the service of industry, government, the engineering profession and society at large, graduates will function effectively in the complex modern work environment with clear communication, responsible teamwork, and high standards of ethics, professionalism, safety and protection of the environment.
4	Life-long learning	Based upon a rigorous undergraduate program that is innovative, challenging, open and supportive, graduates will enhance their skills and knowledge through life-long learning and demonstrate professional leadership.



ChE Curriculum:

Core Courses

Thermodynamics-Based Courses

- ★ ChE 201 – Introduction to thermodynamics
- ★ ChE 301 – Chemical Engineering Thermodynamics
- ChE 321 – Chemical Reaction Engineering

Process-Oriented Courses

- ChE 210 – Material & Energy Balances
- ChE 341 – Chemical Process Control
- ChE 396 – Senior Design I (Equipment Design)
- ChE 397 – Senior Design II (Design teams)

Numerical Analysis

- ChE 205- Numerical Methods in ChE

Engineering Education & Development

- ENGR100 -Engineering Orientation (Engineering Success Program, Guaranteed Paid Internship Program)
- ChE 499- Professional Development Seminar

Transport-Based Courses

- ChE 311 – Transport Phenomena I (Fluid Mechanics)
- ChE 312 – Transport Phenomena II (Heat & Mass Transfer)
- ★ ChE 313 – Transport Phenomena III (Separations)

Unit Operations Laboratory

- ChE 381 – Chem Eng Lab I
- ChE 382 – Chem Eng Lab II

Molecular Engineering Courses

- ★ CHE 230 Molecular Engineering in CHE
- ★ CHE 330 Polymer Science

ChE 400-level Elective
CHE 392:
Undergraduate
Research

Six Concentrations (15% of ChE students):

Biochemical Engineering	9 students
Process Automation	6
Energy & Environment	4
Nanotechnology	3
Polymers & Molecular Engineering	3
Entrepreneurship	0



Curriculum Improvement in progress...

Chemical
engineers are at
the forefront of
transforming our
future with
sustainable
solutions

PROGRAM REVISION IDEAS/ PROPOSALS:

(1)

ADD CHE 101 INTRODUCTION TO SUSTAINABLE
CHEMICAL ENGINEERING TO CORE CURRICULUM (+3
HOURS)

CREATE A NEW SELECTIVE OPTION: CHE 230
MOLECULAR SYSTEMS IN CHEMICAL ENGINEERING
OR CME 260 PROPERTIES OF MATERIALS (-3 HOURS)

(2)

REMOVE CHEMISTRY COURSE FROM MATH/SCIENCE
REQUIREMENT (-3 HOURS)

ADD SECOND TECHNICAL ELECTIVE TO CORE
REQUIREMENT (+3 HOURS)

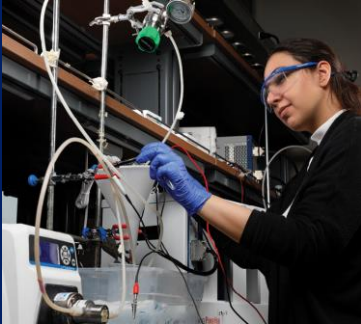
(3)

ADD DATA SCIENCE TO CHE CURRICULUM:

CREATE A NEW DATA SCIENCE CONCENTRATION



JOB OPPORTUNITIES



Chemical Engineering Employment Sectors

biotechnika
Your Bio Resource



Biotechnology



Chemical and allied
products



Process Plant
Construction



Pharmaceutical &
Toiletries



Water Industry



Food & Beverage
industries



Environment Protection
& Recycling



Energy (Oil,
Gas, & Nuclear)



Process plant &
Equipment
Manufacture



Materials (Plastics, dyes,
metals, paper, etc.)



Academics



IT



Consultancy



Others



UIC Chemical Engineering graduates are in high demand



Ana Hernandez

Ana Hernandez (She/Her) · 1st
Process Manager @L'Oréal | People Leader |



Jordan Flanigan · 1st
Semiconductor Process Engineer at Intel



Aesha Talia · 1st
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Emir Asani · 1st
Continuous Improvement Engineer at Mars Ice Cream



Courteney Bruce-Kotey · 1st
Process Engineer at Brindley Engineering Corporation



Paul Diaz · 1st
System Manager for digital I&C and mechanical equipment at Constellation Nuclear



Student - ChE Community Interaction

- WELCOME BACK PICNIC
- END-OF-YEAR BANQUET
- AICHE STUDENT CHAPTER

GENERAL MEETINGS, PROFESSIONAL & FUN ACTIVITIES

AICHE NATIONAL MEETING, NORTH-CENTRAL REGION

CHEME CAR, CHEME CUBE

- CHE EXPO
- UIC CHEMICAL ENGINEERING LINKEDIN GROUP

- PAST EVENTS:

CHE INDUSTRY DAY- PROFESSIONAL ENGINEER PANEL
DISCUSSION

CHE FORUM:

‘MINI-CONFERENCE’ WITH TECHNICAL SESSIONS

ALUMNI ‘MIXER’: STUDENTS & FACULTY INVITED



OCTOBER 1, 2024

Town Hall Meeting: ChE Graduate Program

Brian P. Chaplin

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ChE Graduate Curriculum:

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- 4+1 Format: Two graduate classes senior year
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PhD in Chemical Engineering:

- 6 required core courses
- 6 electives
- Departmental Seminar
- 52-60 semester hours of dissertation research
- Publication requirement set by advisor



Graduate Program Revisions under Consideration...

Proposed Revisions:

- Adding a professional development course
- Reduce number of required courses and increase elective options
- Develop a PhD checklist to self assess progress
- Institute regular formal student assessments by advisor
- Standardize the prelim timing/content
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Discussion Points...

Survey Results:

- **Need for updated graduate student handbook and structured PhD program guidelines**
- **Issues with timely TA/RA letter submissions and funding stability**
- **Disconnect between research labs**
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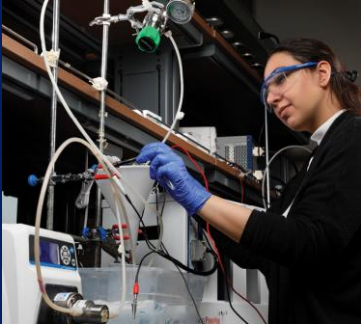
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